

Core Banking Services: Business Requirements

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Owner	Retail Banking Product Office
Approved by	Head of Retail Products; Head of Technology Risk
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Related documents	NBK-NFR-2026-003 (Non-functional requirements), NBK-SEC-2025-021 (Authentication standard)

1. Purpose and scope

This document states the functional requirements for the Accounts, Transfers and Loans services of the Northbridge core banking platform. Every requirement has a unique identifier of the form **REQ-[<MODULE>-<NN>](#)**. Each identifier must be traceable to its implementation and to at least one automated test before a release is approved (see MAS Technology Risk Management Guidelines, section 9).

All monetary amounts are in Singapore Dollars (SGD) and are held to two decimal places. "Calendar day" means a day in Singapore Standard Time (UTC+8). Non-functional requirements (performance, availability, security controls) are out of scope and are covered in NBK-NFR-2026-003.

2. Glossary

Term	Meaning
Active account	An account in status ACTIVE. Accounts may also be FROZEN or CLOSED.
Payee	The destination account of a transfer.
New payee	A destination to which the source account has never completed a transfer.
Instalment	The fixed monthly repayment for an amortising loan.

3. Accounts module

REQ-ACC-01 Account opening

Requirement	A customer can open an account by giving their full name, an initial deposit and a 6-digit PIN.
Acceptance criteria	- A blank or whitespace-only name is rejected. - A negative initial deposit is rejected. An initial deposit of SGD 0.00 is allowed. - A newly opened account has status ACTIVE and a balance equal to the initial deposit.

REQ-ACC-02 Account number format

Requirement	Each account gets an account number made of the prefix "SG" and 8 digits, e.g. SG01234567.
Acceptance criteria	- Account numbers match the pattern <code>SG\d{8}</code>. - No two accounts share an account number.

REQ-ACC-03 Deposits

Requirement	A customer can deposit money into an active account.
Acceptance criteria	- The deposit amount must be greater than zero; zero or negative amounts are rejected. - The balance increases by exactly the deposited amount.

REQ-ACC-04 Withdrawals and insufficient funds

Requirement	A customer can withdraw money from an active account, but never more than the available balance. Overdrafts are not offered.
Acceptance criteria	- A withdrawal larger than the balance is rejected with an insufficient-funds error. - After a rejected withdrawal, the balance is unchanged.

REQ-ACC-05 Account closure

Requirement	An account can be closed only when its balance is exactly SGD 0.00.
Acceptance criteria	- Closing an account with a non-zero balance is rejected, and the account stays ACTIVE. - Closing an account with a zero balance sets its status to CLOSED.

REQ-ACC-06 Frozen accounts

Requirement	An account in status FROZEN cannot receive deposits or make withdrawals.
Acceptance criteria	- A deposit to a frozen account is rejected. - A withdrawal from a frozen account is rejected. - The balance of a frozen account is not changed by rejected operations.

REQ-ACC-07 PIN lockout

Requirement	To prevent brute-force attacks, an account is automatically frozen after 3 consecutive failed PIN verification attempts.
Acceptance criteria	• After the third consecutive incorrect PIN, the account status becomes FROZEN. • A successful verification resets the failed-attempt counter to zero. • Two failures, then a success, then two more failures leaves the account ACTIVE.

REQ-ACC-08 Balance enquiry

Requirement	A balance enquiry returns the account balance rounded to two decimal places (half-up).
Acceptance criteria	• An account holding 10.005 reports a balance of 10.01.

4. Transfers module

REQ-TRF-01 Atomic funds transfer

Requirement	A transfer debits the source account and credits the destination account as one atomic operation.
Acceptance criteria	• After a completed transfer of X, the source balance is lower by X and the destination balance is higher by X. • If the credit cannot be applied (e.g. destination not active), the source balance is unchanged.

REQ-TRF-02 No self-transfers

Requirement	A transfer where source and destination are the same account is rejected.
Acceptance criteria	• Transferring from an account to itself raises a validation error.

REQ-TRF-03 Daily transfer limit

Requirement	The total of outgoing transfers from one account in a calendar day must not exceed SGD 5,000.00. The limit is inclusive : a day's total of exactly SGD 5,000.00 is allowed.
Acceptance criteria	• Given SGD 4,000.00 already transferred today (to an existing payee), a further transfer of SGD 1,000.00 is accepted. • After that, a transfer of SGD 0.01 is rejected with a daily-limit error. • Transfers made on a previous calendar day do not count towards today's limit.

REQ-TRF-04 New payee review

Requirement	A transfer of more than SGD 1,000.00 to a new payee is held for manual fraud review and not executed immediately.
Acceptance criteria	• A first-time transfer of SGD 1,000.01 or more to a payee has status PENDING_REVIEW, and neither balance changes. • A first-time transfer of exactly SGD 1,000.00 completes immediately. • Once one transfer to a payee has completed, later transfers to that payee are not held.

REQ-TRF-05 Transfer reference

Requirement	Every transfer gets a unique reference of the form "TRF-" followed by 8 upper-case alphanumeric characters, plus a timestamp.
Acceptance criteria	• References match TRF-[A-Z0-9]{8} and are unique. • Each transfer records the time it was created.

REQ-TRF-06 Transfer history

Requirement	A customer can view the transfer history of an account, including both incoming and outgoing transfers, most recent first.
Acceptance criteria	• History includes transfers where the account is the source or the destination. • Transfers are listed newest first. • Transfers between two other accounts do not appear.

REQ-TRF-07 Scheduled transfers

Requirement	A customer can schedule a transfer for a future date. The transfer runs automatically on that date and can be cancelled before then.
Acceptance criteria	• A scheduled transfer does not move funds before its execution date. • On the execution date, the transfer is executed under the same rules as REQ-TRF-01 to REQ-TRF-04. • A cancelled scheduled transfer is never executed.

5. Loans module

REQ-LN-01 Loan amount bounds

Requirement	Personal loans can be requested for SGD 1,000.00 to SGD 100,000.00 inclusive.
Acceptance criteria	• 999.99 and 100,000.01 are rejected; 1,000.00 and 100,000.00 are eligible.

REQ-LN-02 Loan tenure

Requirement	The loan tenure is 12 to 60 months inclusive.
Acceptance criteria	11 and 61 months are rejected; 12 and 60 months are eligible.

REQ-LN-03 Monthly instalment calculation

Requirement	The monthly instalment uses the standard amortisation formula $M = P \cdot r \cdot (1+r)^n / ((1+r)^n - 1)$, where r is the annual rate / 12 and n is the tenure in months. The result is rounded to the nearest cent (half-up) .
Acceptance criteria	- SGD 25,000.00 at 5.5% p.a. over 36 months → instalment SGD 754.90. - SGD 60,000.00 at 4.5% p.a. over 48 months → instalment SGD 1,368.21.

REQ-LN-04 Affordability check

Requirement	An application is rejected when the applicant's monthly income is less than 3 times the monthly instalment.
Acceptance criteria	- Income below 3× the instalment → REJECTED with an affordability reason. - Income of 3× the instalment or more → eligible.

REQ-LN-05 Interest rate tiers

Loan amount	Annual rate
Below SGD 20,000.00	6.5%
SGD 20,000.00 to SGD 50,000.00 inclusive	5.5%
Above SGD 50,000.00	4.5%

REQ-LN-06 Early repayment fee

Requirement	A borrower who repays early pays a fee of 2.0% of the outstanding principal, rounded to the nearest cent (half-up).
Acceptance criteria	- Outstanding principal SGD 12,345.67 → fee SGD 246.91. - Outstanding principal SGD 10,000.00 → fee SGD 200.00.

REQ-LN-07 Loan decision

Requirement	Every loan application gets a decision of APPROVED or REJECTED. Rejected applications carry a human-readable reason.
Acceptance criteria	• Approved loans have no rejection reason; rejected loans always have one. • A decided application can be retrieved by its identifier.

6. Change control

Version	Change
1.1	REQ-LN-06: early repayment fee raised from 1.5% to 2.0% (Pricing Committee, 18 Sep 2026).

Changes to this document need Product Office approval. Engineering must re-establish traceability for every changed requirement before the next release.